

BRAIN TUMOR MRI CLASSIFICATION

Deep Learning for Medical Image Analysis | PyTorch | CNN

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89.81%

Improved CNN
Accuracy

87.19%

Basic CNN
Accuracy

4

Tumor
Classes

15

Training
Epochs

Project Overview

This project applies deep learning to a real-world medical problem — classifying brain MRI scans into four categories: glioma, meningioma, pituitary tumor, and no tumor. The model learns to recognize visual patterns in MRI images entirely from data, without any pre-programmed medical knowledge.

Two custom Convolutional Neural Networks (CNNs) were built and compared. The Improved CNN achieved 89.81% test accuracy, outperforming the baseline by 2.62 percentage points through targeted architectural enhancements including batch normalization and an additional convolutional layer.

Why It Matters

Radiologists face growing imaging workloads, and delays in diagnosis directly affect patient outcomes. An AI classification tool can serve as a decision-support layer — flagging scans by tumor type to help prioritize clinical review, without replacing the radiologist's judgment.

Class	Plain-Language Definition	Classification Note
Glioma	Tumor from glial cells — often aggressive, visually complex	Hardest class to classify
Meningioma	Tumor on membranes surrounding the brain — frequently benign	Can resemble Glioma on scans
Pituitary	Tumor on the pituitary gland at the brain's base	Distinct location aids accuracy
No Tumor	Healthy scan — no tumor present	Highest model accuracy

Methodology

The project followed a standard deep learning pipeline for medical image classification:

- Images resized to 128×128 pixels and normalized — consistent preprocessing across all scans
- Batched data loading using PyTorch's DataLoader (batch size 32, GPU-accelerated in Google Colab)

- Both models trained for 15 epochs using the Adam optimizer and Cross-Entropy Loss
- Dropout regularization applied to prevent the model from memorizing training data

Two architectures were evaluated:

	Basic CNN	Improved CNN ✓
Architecture	2 convolutional layers	3 convolutional layers
Batch Norm	None	After each layer ✓
Dropout Rate	0.5	0.3
Test Accuracy	87.19%	89.81%
Verdict	Solid baseline	Best performer

Results & Key Findings

- The Improved CNN outperformed the baseline across all four tumor categories
- No Tumor and Pituitary were the strongest classes — their MRI patterns are most visually distinct
- Glioma and Meningioma were hardest to separate — a known challenge even for trained radiologists
- Batch normalization was the single most impactful change, stabilizing training and reducing overfitting
- Full confusion matrix and per-class precision, recall, and F1-score reported in the project notebook

Business Relevance & Impact

This project demonstrates how deep learning can be applied to high-stakes domains where accurate, explainable AI has real consequences. Building two models and comparing them rigorously reflects the analytical discipline required in production ML — not just building something that works, but understanding why it works and where it falls short.

- A production version of this tool could flag priority MRI scans, reducing time from scan to diagnosis
- The CNN pipeline is transferable to other imaging tasks: chest X-rays, retinal scans, skin lesions
- Future work: transfer learning (ResNet-50), data augmentation, and Grad-CAM for explainability

Tools & Technologies

Component	Details
Language	Python 3
Deep Learning	PyTorch — CNN, Adam optimizer, CrossEntropyLoss
Image Pipeline	torchvision — ImageFolder, transforms, DataLoader
Evaluation	scikit-learn — confusion matrix, classification report
Environment	Google Colab with GPU-accelerated training